sharp margin lesions known as Jock itch. It is caused by *Trichophyton rubrum* and *Epidermophyton floccosum*.

10. **Tinea Manuum:** This is an infection of the **skin of palmar** aspect of hands. It causes hyperkeratosis of the palms and fingers. It is caused by *Trichophyton mentagrophytes*, *Trichophyton rubrum* and *Epidermophyton floccosum*.
11. **Tinea Unguium:** This is an infection of the **nail plates**. The infection spreads on the entire nail plate infecting the nail bed. It results in opaque, chalky or yellowish thick ended nail. It is caused by *Trichophyton mentagrophytes*, *Trichophyton rubrum* and *Epidermophyton floccosum*.

Figure 9.6 shows the microscopic view of major determatophytes

Infobits

How do dermatophytes cause disease in humans?

Dermatophytosis is a common contagious **disease caused** by fungi **known as dermatophytes**. **Dermatophytes** belong to a group of **organisms** that are able to break down the keratin in tissues such as the epidermis, hair, nails, feathers, horns and hooves.

9.2.4 Laboratory Diagnosis

i. Samples

Skin scrapings, hair and nail samples were collected

a. Direct Examination

Samples are subjected to KOH (10%) wet mount, the affected site were disinfected with alcohol before collecting the clinical specimen.

b. Fungal culture

The samples are inoculated on Sabouraud dextrose agar (SDA) with antibiotics and cycloheximide and are incubated at 25°C–35°C. The colony morphology can be identified.

The three genera of dermatophytes are *Trichophyton*, *Microsporum* and *Epidermophyton* (Table 9.1). Based on morphology of the macro conidia, micro conidia, their shape, position on the spore bearing hyphae such as spiral hypha, racquet hypha, nodular pectinate body, they are identified.

ii. Special Techniques

1. Wood's Lamp Examination

Clinical samples are exposed to Wood's lamp. Wood's glass consists of Barium silicate containing 9% Nickel oxide. It transmits long wave ultra violet light with a peak of 365nm that shows a characteristic fluorescence produced by the samples. The patterns of fluorescence are bright green, golden yellow and coral red. *Microsporum* species and *Trichophyton* species are differentiated using this technique.

2. Hair brush sampling Technique

It involves brushing the scalp with a sterile plastic hair brush, which is then inoculated into an appropriate culture medium by plates, is incubated at 25°C–35°C. The colony morphology can be identified.

TN_GOVT_XII_Micro_Biology_CH09.indd 195 27-02-2019 14:55:16